KATYANI SINGH

PhD Researcher in Computing Science, University of Alberta

Multimodal AI · Fairness & Robustness · Interpretability · AI Alignment

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SUMMARY

PhD researcher in Computing Science with **5+ years of experience** in machine learning and computer vision, spanning **2.5+ years in industry** as a Machine Learning / Computer Vision Scientist (Nextech3D.ai) and Computer Vision Researcher at AI startups, and **3+ years of graduate research** (MSc and PhD) at the University of Alberta. Built and deployed production systems with latent diffusion models, NeRF, vision-language models and contrastive retrieval.

EDUCATION

Doctor of Philosophy, Computing Science **2025 – Present**
University of Alberta, Edmonton AB *GPA: 4.0/4.0*
Supervisor: Dr. Nidhi Hegde
Thesis Focus: Fairness, Optimization, Robustness, and Transparency in Vision-Language Models
Academic Highlights: **4.0/4.0 GPA** with an **A+ in every graduate course**. recipient of **three nomination-only graduate scholarships** in 2026, each awarded to a single student per year for superior academic achievement.

Coursework:

- Reinforcement Learning I (CMPUT 655) **A+**
- Deep Reinforcement Learning (CMPUT 628) **A+**
- Topics in Computing Science (CMPUT 605) **A+**

Master of Science, Computing Science (Thesis-based) **2021 – 2023**
University of Alberta, Edmonton AB *GPA: 3.9/4.0*
Supervisor: Dr. Nilanjan Ray
Thesis: Unsupervised Document Image Denoising for OCR
Research Focus: Unpaired Document Image-translation using Generative Adversarial Networks
Thesis research funded by Intuit Inc. and NSERC; first-author journal publication in IJDAR (2024).

Bachelor of Technology, Computer Engineering **2016 – 2020**
MPSTME, NMIMS University, India *Rank: 2/120*
Graduated **2nd in a class of 120**. published first-author IEEE conference paper as an undergraduate.

AWARDS AND HONOURS

Verna Tate Graduate Scholarship in Science **$11,500** **Oct 2026**
Faculty of Graduate and Postdoctoral Studies, University of Alberta *2026 – 2027*
Endowed scholarship awarded to **only one** female graduate student per year in the Department of Computing Science on the basis of superior academic achievement.

Maya and Mangesh Murdeshwar Graduate Award **$1,900** **Oct 2026**
Faculty of Graduate and Postdoctoral Studies, University of Alberta *2026 – 2027*

Awarded to **a single** female student per year, selected on academic achievement in Computing Science department.

Ashley Finlayson and Diane Hollinshead International Graduate Scholarship in Science **$1,500** **Sept 2026**
Faculty of Graduate and Postdoctoral Studies, University of Alberta *2026 – 2027*
Awarded to **one** international graduate female student per year across the Departments of Computing Science, Mathematical & Statistical Sciences, and Physics for superior academic achievement.

Silicon Valley Fellowship **2026**
Silicon Valley Fellowship, San Francisco, CA *Sept 2026*
Selected for a week-long immersion program with leaders, founders, and researchers from Big Tech and top universities in the Bay Area. **∼2% acceptance rate** from **5,000+ applications** per year.

Gold Medal, Rank 2 in Department **2020**
MPSTME, SVKM's Narsee Monjee Institute of Management Studies (NMIMS)
Awarded a **gold medal** for graduating **2nd out of 120 students** in the Computer Engineering Department.

Dean's Meritorious Students List **Aug 2020**
MPSTME, SVKM's Narsee Monjee Institute of Management Studies (NMIMS)
Named to the Dean's list for academic merit; graduated with 2nd rank in the Computer Engineering Department.

Student Ambassador Award **Aug 2020**
MPSTME, SVKM's Narsee Monjee Institute of Management Studies (NMIMS)
Recognized by the institute at graduation for representing NMIMS as a student ambassador, combining academic excellence with leadership and contributions to the university community.

Winner, Smart India Hackathon **$1,500** **March 2019**
Developed autonomous invoice processing platform using RPA and deep learning; selected as **top solution among 10,000+ participants** nationwide. The solution is under incubation by the Department of Science and Technology, Government of India.

INDUSTRY EXPERIENCE

AI Career Accelerator Participant (Project Validation) **Sept 2026 – Present**
Alberta Machine Intelligence Institute (AMII)

- Supporting the AMII Industry Team in delivering Project Validations, which help client companies scope a business problem into an executable ML project definition, identify viable ML approaches, and understand the associated risks.
- Conducting academic and industrial literature reviews and brainstorming ML approaches to solve each client's business problem.
- Participating in client meetings and creating slides and refined documentation of findings and approaches that are integrated into the final client report.

AI Career Accelerator Participant (Client Coaching) **Sept 2026 – Present**
Alberta Machine Intelligence Institute (AMII)

- Collaborating with AMII startup team ML scientists as domain expert, consulting startups through ML adoption journeys including problem identification, evaluation, and proof-of-concept planning and execution.

Machine Learning / Computer Vision Scientist **2023 – 2025**
Nextech3D.ai

- Architected text-to-image virtual photography pipeline using latent diffusion models with custom relighting, achieving photorealistic 4K product integration for major retail clients.
- Designed image-to-3D reconstruction system combining diffusion priors with NeRF, enabling high-fidelity 3D asset generation from single images.
- Built multimodal retrieval systems for 3D assets using contrastive learning, improving search accuracy by **35%**.
- Deployed vision-language models (BLIP, InstructBLIP) for automated product metadata generation, reducing manual annotation time by **80%** with **92% accuracy**.

RESEARCH EXPERIENCE

Graduate Research Assistantship Fellow (PhD) **2025 – Present**
University of Alberta, Department of Computing Science *Edmonton, AB*
Supervisor: Dr. Nidhi Hegde

- Conducting doctoral research on fairness, robustness, and transparency in vision-language models.
- Developing principled debiasing methods leveraging distributionally robust optimization (DRO) across multimodal architectures.
- Within the first 12 months of the PhD: co-authored an **oral presentation at ACM FAccT 2026**, first-authored a **NeurIPS 2026 workshop paper**, and was selected as a reviewer for ACM FAccT 2027.

Graduate Research Assistant (MSc) **May 2021 – July 2023**
University of Alberta, Department of Computing Science *Edmonton, AB*
Supervisor: Dr. Nilanjan Ray

- Conducted thesis research on unsupervised document image enhancement for OCR, funded by Intuit Inc. and NSERC.
- Developed novel framework for unpaired image-to-image translation, achieving **state-of-the-art** results with **16.39% OCR improvement**. Published in IJDAR 2024.

Computer Vision Researcher **Aug 2020 – Dec 2020**
Eigenlytics Data Solutions Pvt. Ltd. *India*

- Designed end-to-end OCR system for financial documents, improving text localization accuracy by **28%** through adaptive morphological operations and layout-aware segmentation.

Computer Vision Researcher **April 2019 – Feb 2020**
iPing Data Labs LLP *India*

- Proposed a novel model for defect detection in thermal insulation materials, enhancing bubble

detection accuracy by **32%** through adaptive filtering.

Machine Learning Researcher **May 2019 – June 2019**
Ytrre Software Solutions Pvt. Ltd. *India*

- Developed multimodal document understanding framework using LayoutLM-based classification, achieving **89%** classification accuracy on heterogeneous document corpora.

PUBLICATIONS

Singh K, Hegde N. Nonlinear Bias Survival after Post-Hoc Debiasing of Vision-Language Representations. *AI4Good Workshop at the Conference on Neural Information Processing Systems (NeurIPS)*, 2026. **Accepted poster.**

Gandhi D, **Singh K**, Hegde N. RobustDebias: Debiasing Language Models using Distributionally Robust Optimization. *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2026. **Oral presentation.** [Link]

Singh K, Ray N. Unpaired Document Image Denoising for OCR using BiLSTM enhanced Cycle-GAN. *International Journal on Document Analysis and Recognition (IJDAR)*, 2024. [Link]

Tata G, **Singh K**, Oeveren E, Ray N., Document Image Cleaning using Budget-Aware Black-Box Approximation. *arXiv*, 2023. [Link]

Singh K et al. A Comprehensive Review of Convolutional Neural Network based Image Enhancement Techniques. *IEEE ICSCAN*, 2019. [Link]

PRESENTATIONS AND WORKSHOPS

Selected Poster Presenter, AWS Academic Poster Session **2026**
Grace Hopper Celebration (GHC)
Poster: *Post-hoc AI Debiasing Still Fails Women and Marginalized Groups*

Selected Poster Presenter, AI4Good Workshop **2026**
Conference on Neural Information Processing Systems (NeurIPS)
Poster: *Nonlinear Bias Survival after Post-Hoc Debiasing of Vision-Language Representations*

TEACHING EXPERIENCE

Teaching Assistant, CMPUT 200: Ethics of Data Science and AI **Winter 2025, Fall 2026**
University of Alberta, Department of Computing Science *Edmonton, AB*

Teaching Assistant, CMPUT 566: Introduction to Machine Learning **Fall 2025**
University of Alberta, Department of Computing Science *Edmonton, AB*

Teaching Assistant, CMPUT 175: Foundations of Computation 2 **Winter 2022**
University of Alberta, Department of Computing Science *Edmonton, AB*

Teaching Assistant, CMPUT 101: Introduction to Computing **Fall 2021, Fall 2022**
University of Alberta, Department of Computing Science *Edmonton, AB*

SCHOLARLY SERVICE

Selected Reviewer **2027**
ACM Conference on Fairness, Accountability, and Transparency (FAccT)

Selected Reviewer **2026**
Trustworthy AI for Good Workshop, Neural Information Processing Systems (NeurIPS)

RESEARCH PROJECTS

Geometric Flow for Nonlinear Debiasing of Vision-Language Embeddings *Research in Progress*

Developing a geometric flow framework for debiasing CLIP embeddings by modeling demographic bias as a differentiable scalar field over the embedding manifold. Inspired by score-based diffusion and classifier guidance, the proposed method continuously transports embeddings toward fair regions while preserving semantic structure and manifold geometry.

Neural Ensembles for Contextual Bandits *Research in Progress*

Proposed NE-TS and NE-UCB, novel algorithms leveraging ensemble neural networks with bootstrap resampling for uncertainty quantification in sequential decision-making. Comprehensive empirical evaluation characterized computational-performance tradeoffs for contextual bandit applications.

Fairness-Aware Synthetic Healthcare Data Generation

Formulated GAN-based framework for generating synthetic tabular healthcare data with provable fairness guarantees through multi-objective optimization. Demonstrated **40% reduction** in demographic disparity while maintaining **95%** downstream task performance.

Privacy-Utility-Fairness Tradeoffs in Synthetic Data

Conducted systematic empirical investigation comparing Tabular GAN, CTGAN, and PATE-GAN, characterizing fundamental tradeoffs between differential privacy guarantees, utility preservation, and fairness metrics.

Self-Supervised Learning for Brain MEG Signal Representation

Extended BENDR architecture with novel pretraining objectives combining Contrastive Predictive Coding (CPC) and masked language modeling. Demonstrated **15% improvement** in downstream classification tasks on limited labeled MEG data.

PROFESSIONAL DEVELOPMENT

AI Career Accelerator Participant **May 2022 – Oct 2022**
Alberta Machine Intelligence Institute (AMII)

- Collaborated with AMII startup team ML scientists as domain expert, consulting startups through ML adoption journeys including problem identification, evaluation, and proof-of-concept planning and execution.

CIFAR Deep Learning + Reinforcement Learning (DLRL) Summer School July 2025
Canadian Institute for Advanced Research (CIFAR)

- Selected for CIFAR's flagship AI training program, which reviews **thousands of applications** each year for a cohort of roughly **150–200 trainees** from around the world; lectures and mentorship from leading researchers in deep learning and reinforcement learning.

TECHNICAL SKILLS

Programming	Python, C++, SQL, Bash
Deep Learning	PyTorch, PyTorch Lightning, JAX, TensorFlow, CUDA, PyTorch3D, Open3D
GenAI & LLMs	Hugging Face (Transformers, Diffusers, PEFT, Accelerate), LoRA/QLoRA fine-tuning, RLHF/DPO, vLLM, LangChain, LlamaIndex, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Agents, OpenAI & Anthropic APIs
Multimodal & Vision	CLIP, BLIP/InstructBLIP, LLaVA, Stable Diffusion, ControlNet, NeRF, Gaussian Splatting, OpenCV
Responsible AI	Fairness Metrics, Bias Auditing, DRO, Fairlearn, AIF360, Differential Privacy, Model Explainability (SHAP)
Data & Retrieval	NumPy, Pandas, SciPy, Scikit-learn
MLOps & Deployment	Docker, Kubernetes, MLflow, Weights & Biases, FastAPI, Gradio, Streamlit, GitHub Actions
Cloud	AWS (SageMaker, Bedrock), Google Cloud Platform (Vertex AI), Azure ML